Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

R26-CS-010

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Comments for the Proposal

Comment	Solutions
IT22110220 (Utility Evaluation) and **IT22541284 (Dataset Generation)** did not have dedicated security components in their individual parts. The panel noted that a Cyber Security specialization project requires each member to demonstrate a clear security contribution.	• **IT22110220** implemented a **Data Poisoning Detection** mechanism that runs statistical checks on label distributions across decrypted dataset batches before model training, • **IT22541284** integrated **AES-256** encryption, **SHA-256** integrity hashing, and **RBAC access control** into the dataset generation pipeline.

THE PROBLEM

- AV perception models require millions of labelled sensor frames. Real driving footage captures faces, plates and daily routes (personal data under GDPR and Sri Lanka PDPA)
- Simulation platforms reduce real-data use but calibrate from real records, reintroducing privacy risk through the calibration channel.
- No existing AV simulator provides a formal privacy guarantee or protects the calibration step.

Impact and Gravity

EUR 1.6B
GDPR fines across EU member states in 2022

USD 556B
Projected global AV market by 2026

Individuals, the AV industry, the research community, and society all affected by the absence of privacy-safe AV training data.

Design Excellence

A privacy-by-design pipeline that generates synthetic AV training data with formal epsilon-DP guarantees.

Epsilon-DP enforced at both stages:
Calibration: Laplace noise on Waymo aggregate stats before CARLA.
Training: Opacus DP-SGD with RDP budget accountant.

Privacy guarantee is measured and validated, not assumed.

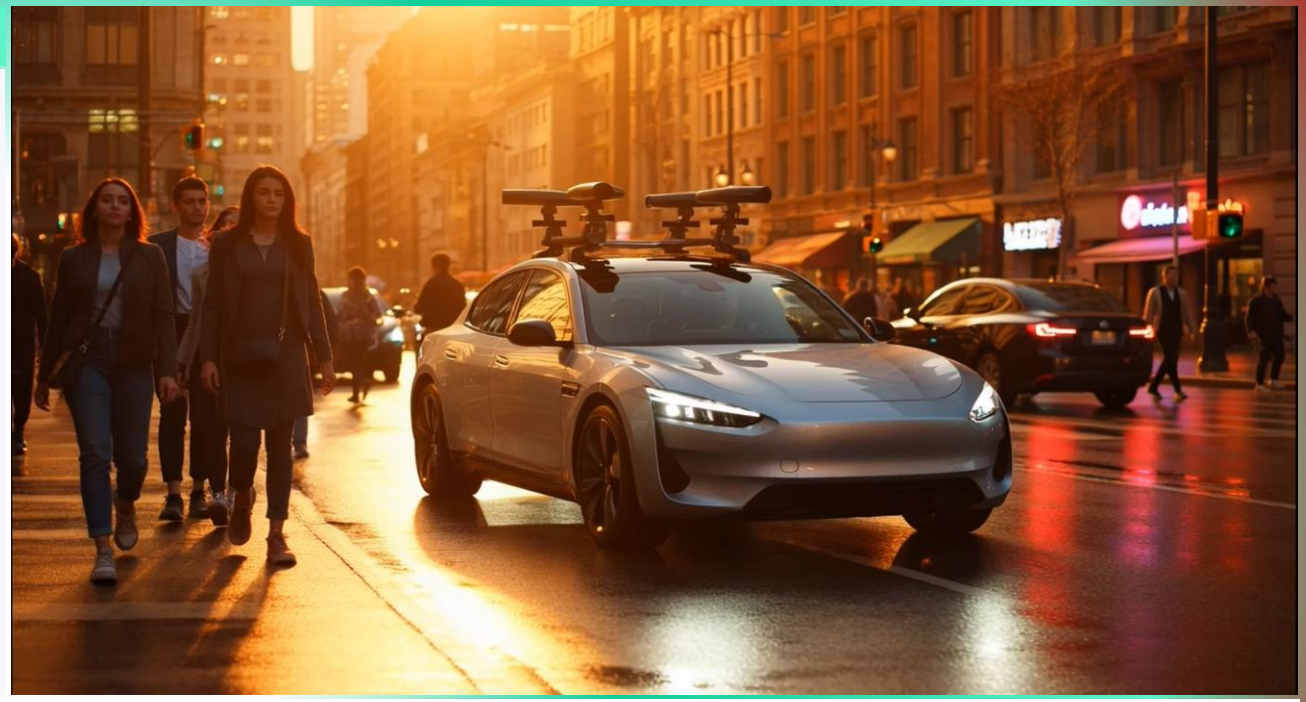
THE PROBLEM

Existing Solutions and Why They Fail

Face and plate blurring: does not protect against membership inference at the model level.

CARLA / AirSim / NVIDIA DRIVE Sim: no formal privacy certificate, calibration channel unprotected.

DP-GAN / DP-CGAN: validated on MNIST only, not multi-modal AV sensor data.



Who Benefits:

AV Developers: GDPR-compliant training data with auditable epsilon-DP certificate

Pedestrians and Road Users: protected from PII exposure in AI training pipelines

Data Protection Regulators: verifiable audit report to confirm lawful data processing

Academic Researchers: shareable privacy-safe AV datasets for collaborative study



IT22309556 | S. A. S. D. PRIYADARSHI

Individual Component: **Privacy Integration**

Novelty:

- Privacy enforced during data generation rather than post-processing.

PRIVACY INTEGRATION

What I Build (Phase 1 - 2)

- **config.yaml**
All project settings in one place (epsilon values, epochs, paths, attack parameters)

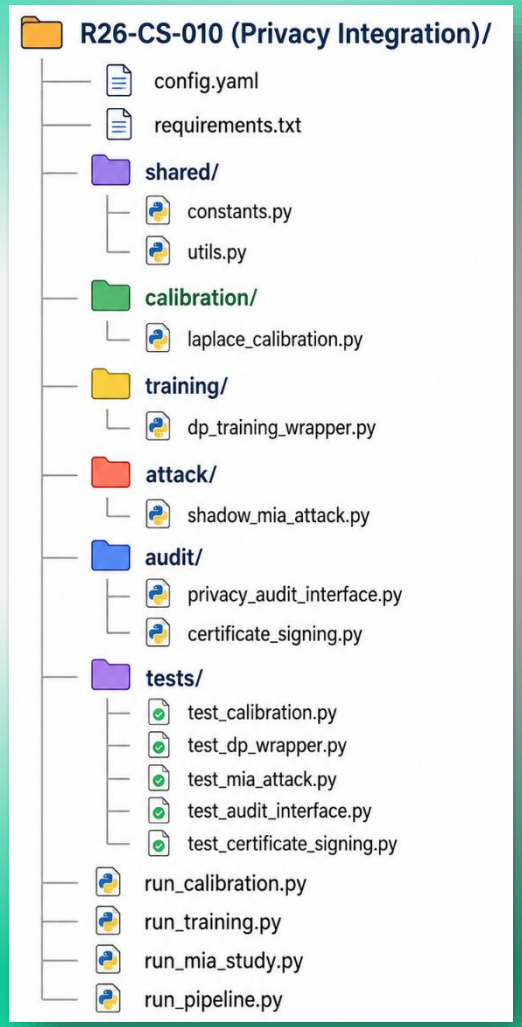
- **requirements.txt**
Python libraries needed to run the project.

- **run_calibration.py**
CLI entry point. Loads a raw calibration JSON, runs the Laplace module, saves calibration_stats.json and calibration_audit_log.json

- **run_training.py**
CLI entry point. Runs DP training across all three ablation configs. Saves budget logs, audit reports, and signs each report with HMAC-SHA256.

- **run_mia_study.py**
CLI entry point. Trains target models for each config and runs the full shadow-model MIA calibration study. Saves calibration_curve_data.json.

- **run_pipeline.py**
CLI entry point. Chains all three runners in sequence as a single end-to-end pipeline command.



- **laplace_calibration.py**
Laplace noise applies to real Waymo aggregate statistics before they reach the CARLA simulator. Raw sensor data never enters this module.

- **dp_training_wrapper.py**
Wraps any PyTorch model with Opacus to enforce epsilon-DP during training via per-sample gradient clipping and Gaussian noise injection.

- **shadow_mia_attack.py**
Shadow-model membership inference attack with two-feature signal (confidence + loss). Produces the empirical epsilon-to-attack-resistance calibration curve.

- **privacy_audit_interface.py**
Packages training results into a structured audit report and privacy certificate delivered to IT22066916 before MIA testing begins.

- **certificate_signing.py**
Sign every audit report with HMAC-SHA256 to create a tamper-evident record. Verification fails if the report is modified after signing. Satisfies GDPR Article 5 accountability.

PRIVACY INTEGRATION

How the Solution Addresses the Sub-Problem and User Requirements

How the Solution Addresses the Sub-Problem

- **Calibration Channel - Laplace Mechanism**

AV simulators calibrated from real driving records reintroduce privacy risk. The Privacy-Aware Calibration Module reads aggregate Waymo statistics in memory, applies Laplace noise (sensitivity / epsilon_calibration), and passes only the privatized values to CARLA. Raw sensor frames are never written to disk.

- **Training Channel - Opacus DP-SGD**

Generative models calibrated on real data can retain statistical traces of individuals. The DP Training Wrapper clips per-sample gradients to L2 norm C = 1.0 and injects Gaussian noise (C x sigma) at every step via Opacus, bounding what any adversary can learn from the model weights.

- **Budget Enforcement - RDP Accountant**

Without automatic budget control, epsilon can be exceeded undetected across a long training run. The Privacy Budget Accountant uses Opacus's built-in RDP composition to accumulate epsilon cost per step and raises BudgetExceededError, halting training when cumulative epsilon reaches 10.0.

- **Audit Trail - Privacy Audit Interface**

No existing AV synthetic data platform produces a verifiable privacy record. The Privacy Audit Interface packages final epsilon, clipping norm C, noise multiplier sigma, and the per-step budget_log.csv into a structured audit_report.json delivered to IT22066916 before MIA tests begin.

User Requirements Addressed

FR-01	Every training run enforces epsilon-DP. RDP accountant log confirms ceiling not exceeded.
FR-02	Per-sample gradient clipped to L2 norm C = 1.0. No sample exceeds C in gradient logs.
FR-03	Gaussian noise (C x sigma) injected at every step before the optimizer weight update.
FR-04	RDP accountant halts training automatically when cumulative epsilon reaches 10.0.
FR-05	Laplace noise applied to Waymo aggregate stats before CARLA. No raw frames in outputs.
FR-06	Audit report (epsilon, C, sigma, total steps) written automatically at close of each run.
FR-07	Per-step epsilon exported as CSV: columns step, cumulative_eps, loss.
NFR-01	Training overhead below 30% vs baseline. Memory footprint below 2x at batch size 32.

PRIVACY INTEGRATION

Design Excellence and Prototype Results

Design Excellence and Novel Contribution

- **Epsilon Calibration Study (Novel Research Contribution)**

Every DP paper reports formal epsilon as evidence of privacy. No prior work tests whether epsilon actually predicts real-world attack resistance for AV data.
This component produces the first empirical epsilon-to-attack-resistance calibration curve for multi-modal AV synthetic data, running the shadow-model MIA against models trained at epsilon = 1.0, 3.0, 8.0, and a no-DP baseline.

- **Extension Beyond Shokri et al. (2017)**

Standard MIA uses confidence score as the sole attack signal. This implementation adds per-sample loss as a second feature, because per-sample loss captures gradient-level memorization more directly in generative architectures. Attack classifier input extended from 1 to 2 features.

User Feedback:

IT22541284 confirmed calibration_stats.json loaded correctly into CARLA.

Prototype Results (Phase 1-2 Validated)

Three ablation configurations: RTX 3050 Ti CUDA, Opacus 1.4.1, PyTorch 2.5.1

Config	Target eps	Final eps	Sigma	Steps	Budget OK
Config A	1.0	0.9921	2.5977	250	True
Config B	3.0	2.9988	1.1938	250	True
Config C	8.0	7.9942	0.7498	250	True

- **Calibration Module**

7 aggregate fields privatized. calibration_stats.json delivered to IT22541284. Calibration eps = 1.8, within the 20% budget cap of 2.0.

- **DP Training Wrapper**

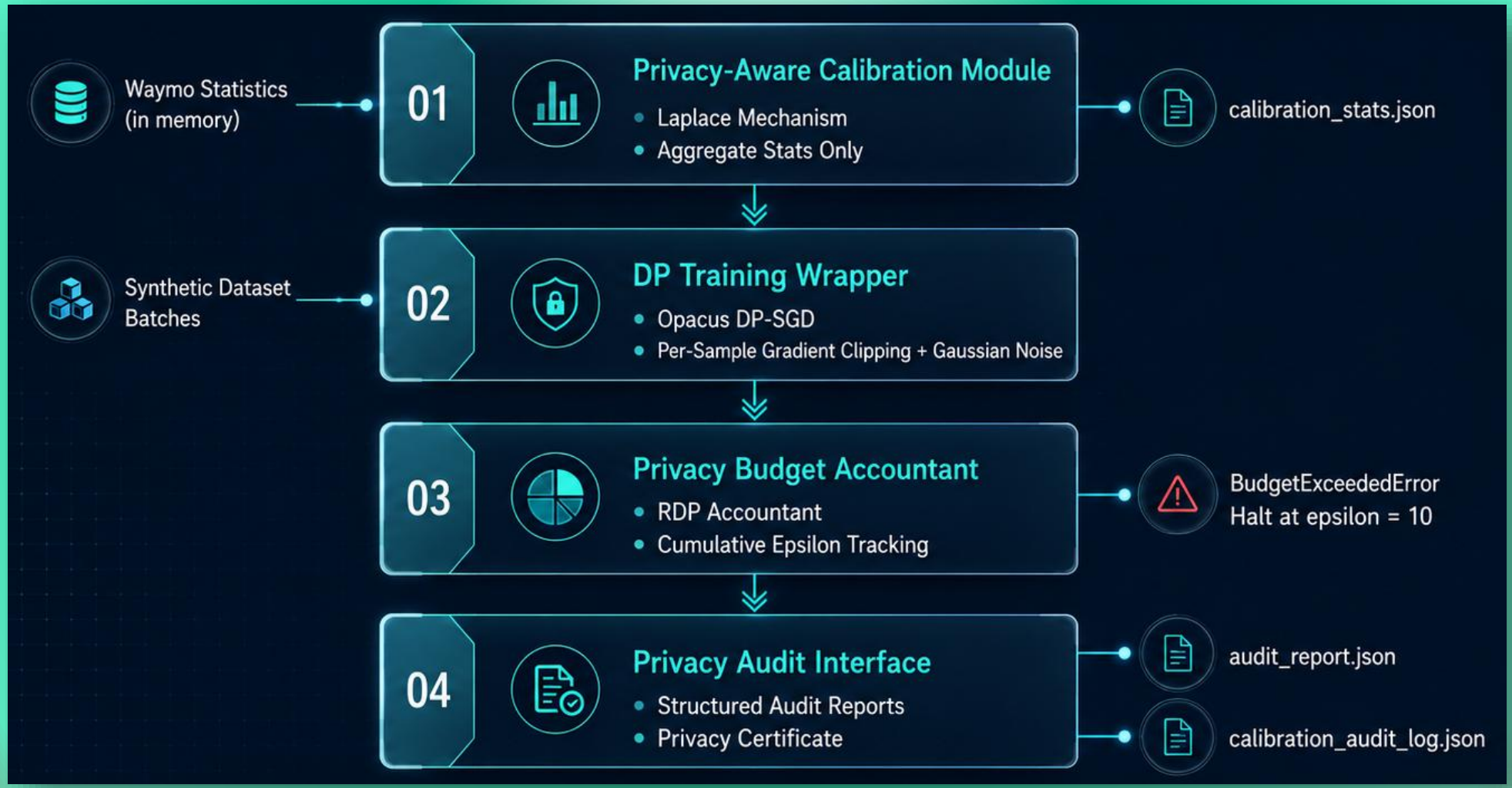
All 3 configs: halted_early = False, budget_respected = True. FR-01 through FR-07 all confirmed. Per-step budget_log.csv exported.

- **MIA Attack Framework**

End-to-end validation complete. Attack success rate 50% on toy data (expected real calibration curve requires Phase 3 CARLA model). Framework ready for Phase 3.

PRIVACY INTEGRATION

Overall Architecture





IT22541284 | YUBITHA S.

Individual Component: **Secured Dataset Generation**

Novelty:

- Generation of **privacy-aware synthetic multi-sensor datasets directly from the simulator environment**, enabling realistic AV training data while preserving privacy constraints.

Integrated Cybersecurity Layer

How Solution Addresses Sub-Problem

Synthetic data eliminates real pedestrian identities, vehicle plates & location traces. No real-world data ever enters the pipeline.

SHA-256 hashing creates a unique digital fingerprint per image batch, detecting tampering or corruption before any model training begins.

AES-256 encryption protects all dataset files at rest. Even if files are copied, they cannot be opened without the correct decryption key

User Requirements Addressed

- Dataset must be protected before sharing with ML researchers.
- Unauthorized access to raw training data must be prevented.
- Researchers need to verify data integrity before use.
- Access must be controlled by role (admin vs researcher).

Design Excellence & Contribution

Embeds security at generation time, not as a post-processing step

Structured metadata manifests enable batch-level traceability

RBAC only authorized roles can decrypt and access datasets

DATA ACQUISITION

This research utilizes a dual-source dataset to ensure high-fidelity training and real-world robustness.

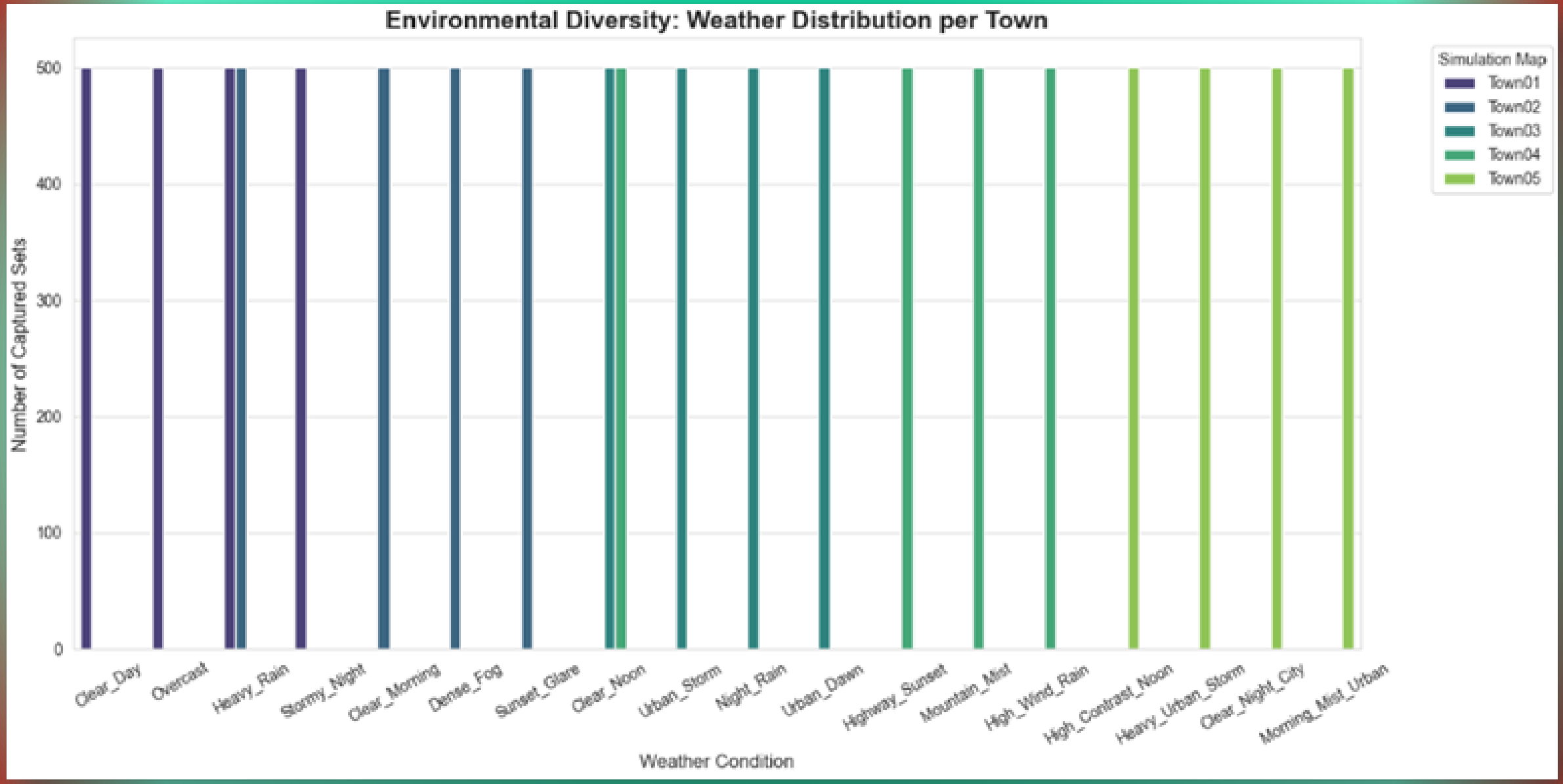
Digital Twin Simulation (CARLA):

- 14-Hour Simulation: Captured 10000 images various .
- Utilized for safety validation by global industry leaders, including Mercedes-Benz, Toyota, NVIDIA, and major Chinese automotive firms.

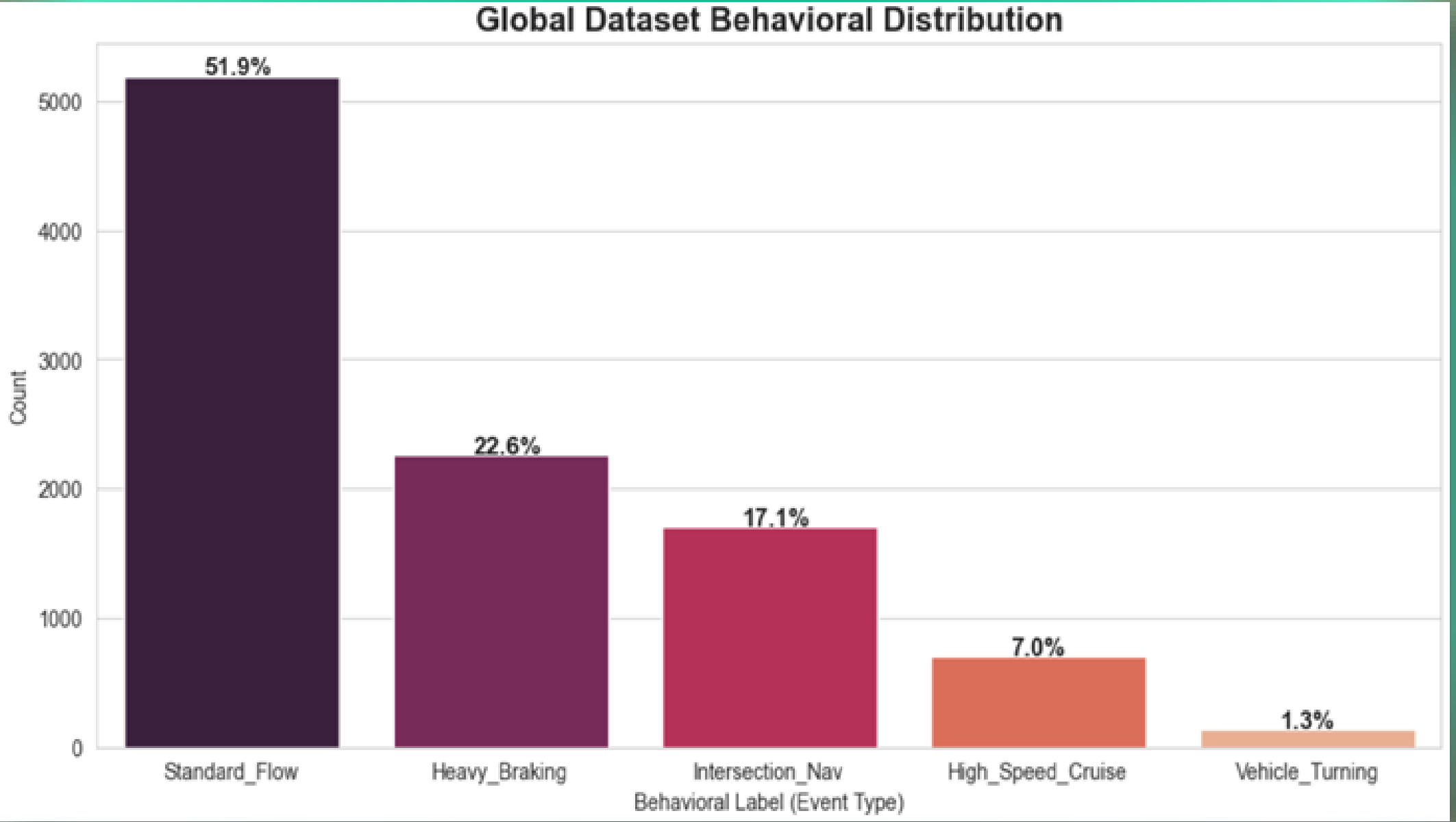
Data Gathering : The Hybrid Dataset Approach

Town	Environment Type	Target Data Records / Sets	RGB Images	Depth Images	Total Image Files	Metadata Rows	Manifest Hash Records	Weather Conditions	Traffic Vehicles	Walkers	Event Types
Town01	Basic urban town	2,000	2,000	2,000	4,000	2,000	4,000	4	35	20 spawned	Intersection_Nav, Vehicle_Turning, Heavy_Braking, Standard_Flow, Unknown
Town02	Quiet residential	2,000	2,000	2,000	4,000	2,000	4,000	4	15	35 configured	Intersection_Nav, Vehicle_Turning, Heavy_Braking, Standard_Flow, Unknown
Town03	Urban city	2,000	2,000	2,000	4,000	2,000	4,000	4	40	25 configured	Intersection_Nav, Vehicle_Turning, Heavy_Braking, Standard_Flow, Unknown
Town04	Highway / high-speed	2,000	2,000	2,000	4,000	2,000	4,000	4	45	10 configured	Intersection_Nav, Vehicle_Turning, Heavy_Braking, High_Speed_Cruise, Standard_Flow, Unknown
Town05	Urban + highway hybrid	2,000	2,000	2,000	4,000	2,000	4,000	4	50	30 configured	Intersection_Nav, Vehicle_Turning, Heavy_Braking, High_Speed_Flow, Standard_Flow, Unknown

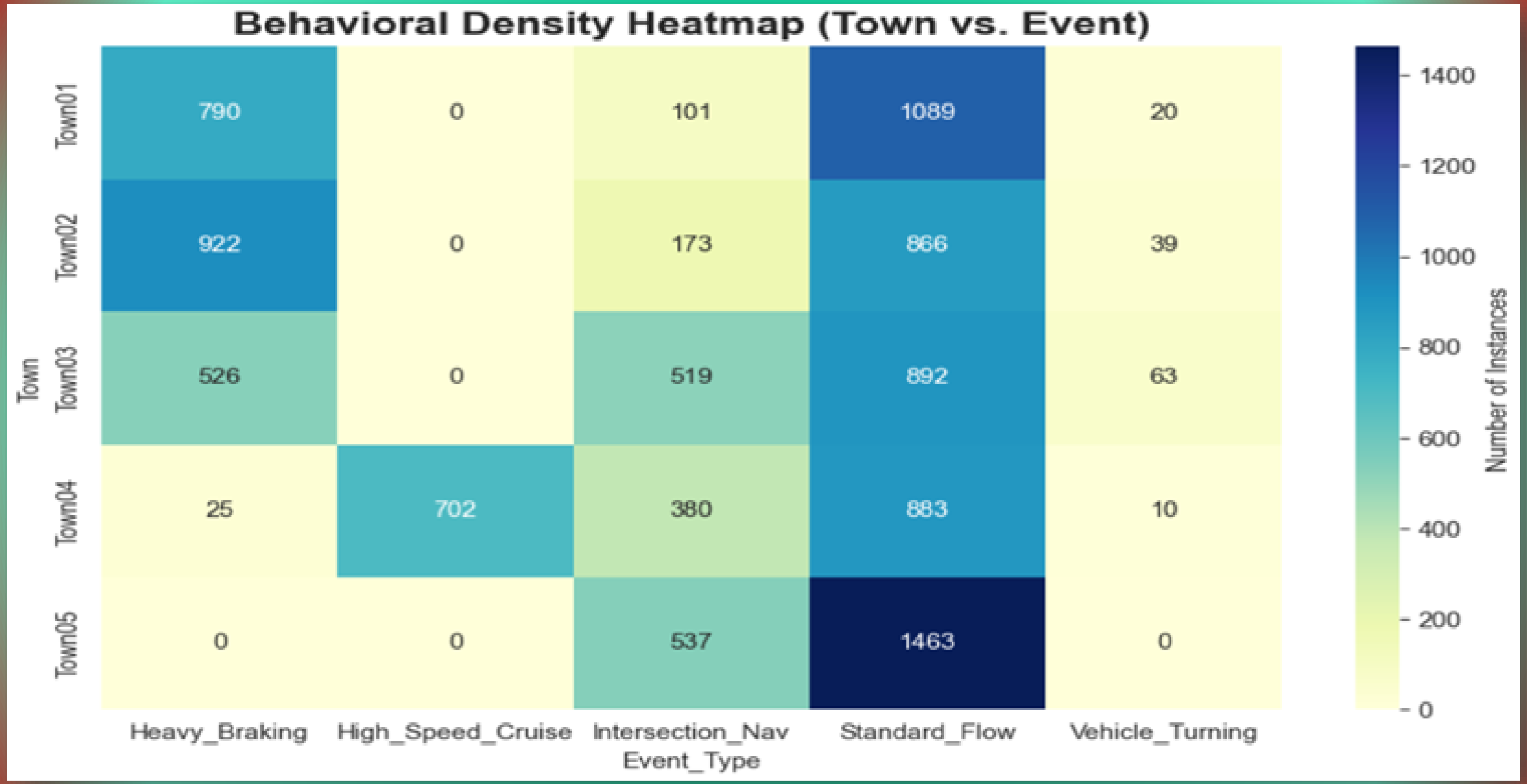
Analysis Report



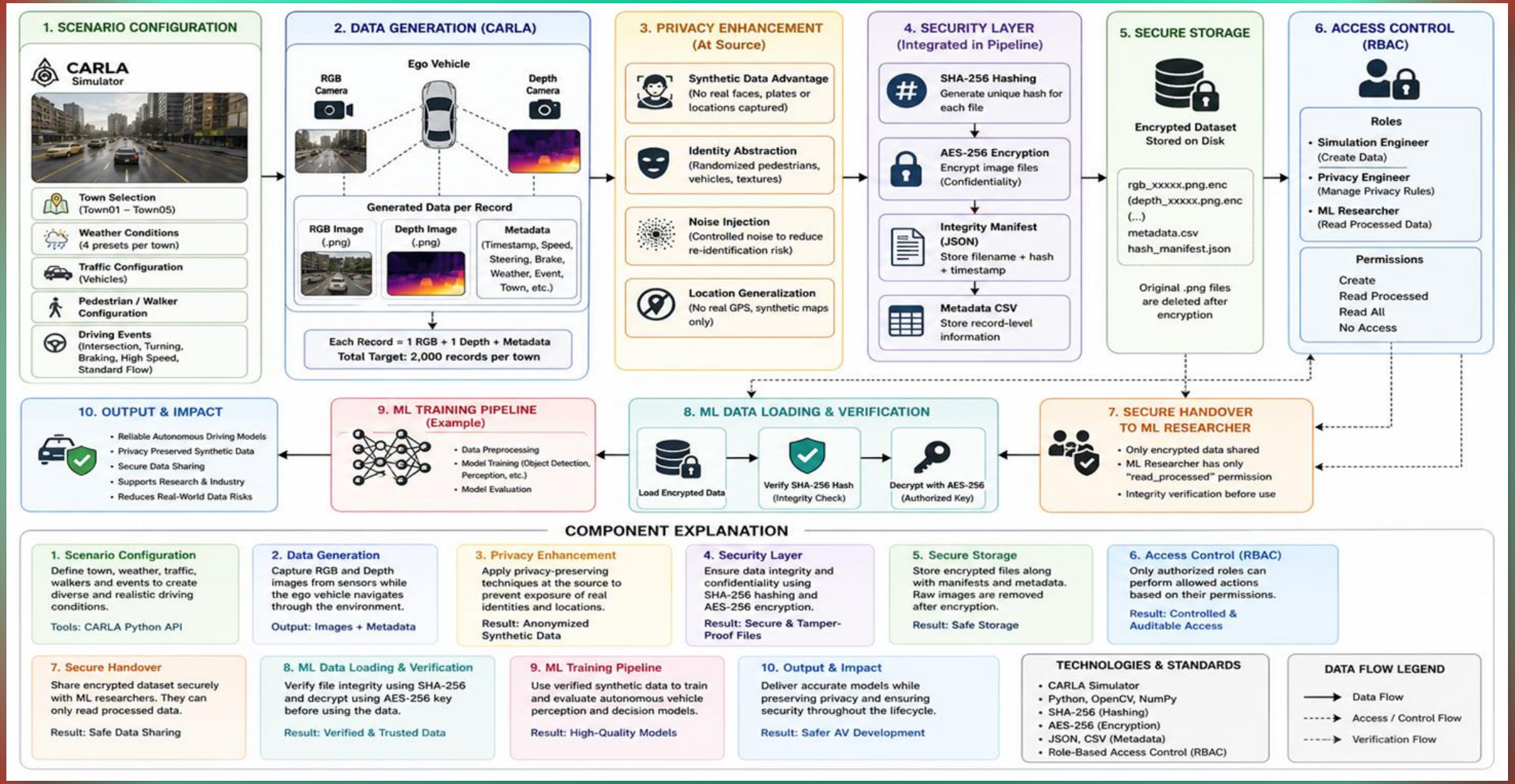
Analysis Report



Analysis Report



Overall Architecture





IT22110220 | M.K.S.R. APEKSHA

Individual Component: **Utility Evaluation**

Novelty:

- Framework generating privacy–enhanced synthetic datasets evaluating AV perception privacy–utility trade–off.

PHASE 1 - BASELINE UTILITY METRICS SETUP

What Was Done

- Set up **PyTorch** deep learning framework in Python 3.12.13 environment with all required dependencies.
- Selected **YOLOv8** (You Only Look Once) as the pre-trained baseline AV perception model for object detection.
- Downloaded and prepared **Waymo** real-world driving dataset as the baseline evaluation dataset.
- Ran the pre-trained **YOLO** model on **Waymo** dataset sample to generate baseline predictions.
- Established baseline utility metrics: **mAP, Precision, Recall** and **F1 Score** on real-world data

Baseline Metrics Established

mAP	**Mean Average Precision** Overall detection accuracy
Precision	**Correct detections/ Total detections** False positive measure
Recall	**Detected objects / All actual objects** False negative measure
F1 Score	**Harmonic mean of Precision & Recall** Balanced accuracy score
FID	**Fréchet Inception Distance** Synthetic data realism score

PHASE 2 - AUTOMATED GROUND-TRUTH LABEL EXTRACTION PIPELINE

01	02	03	04	05
CARLA Simulator	**Sensor Setup**	**Sync Capture**	**Label Extraction**	**Label Alignment**
Connected to CARLA server. Verified world loaded with 35 vehicles & 20 pedestrians in Town01	Attached RGB Camera + Depth Camera to ego vehicle (Tesla Model 3) in synchronous mode at 20 FPS	Captured 10 synchronized frame pairs. All frames saved with matching IDs ensuring temporal alignment	Extracted bounding boxes from simulator's ground truth actor positions using 3D-to-2D projection	Ensured all labels remain perfectly aligned with privacy-altered sensor data from IT22309556

PHASE 2 - DECRYPTING ENCRYPTED SENSOR FILES FROM CARLA SIMULATOR

Encryption Context

- **Source:**
IT22541284 provided output.zip containing CARLA sensor data across 5 town folders (town01–town05)

- **File Types:**
.png.enc (encrypted) and .png (normal) files inside each town folder

- **Algorithm:**
AES-256 CBC Mode identified from security_engine.py provided by IT22541284

- **Key:**
b'ResearchKey_IT22541284_CyberSec!'
32-byte key hardcoded in security_engine.py

- **IV Location:**
First 16 bytes of every .enc file store the Initialization Vector (IV)

- **Library:**
pycryptodome installed in av_privacy_env

Decryption Process

1. Installed **pycryptodome** library in av_privacy_env conda environment
2. Analyzed **security_engine.py** to identify AES-256 CBC encryption method and key
3. Built **decrypt_all_images.py** that automated script to walk all 5 town folders
4. Read first 16 bytes of each .enc file as IV, remaining bytes as ciphertext
5. Applied AES.new (KEY, AES.MODE_CBC, iv) and unpad () to restore original PNG bytes
6. Saved decrypted RGB, Depth images to /decrypted/ folder

HOW WILL THE SOLUTION ADDRESS THE SUB PROBLEM AND USER REQUIREMENTS

Sub Problem Addressed

Quantifying the "Privacy-Utility" gap. The solution measures the accuracy decay of perception models when trained on encrypted/obfuscated synthetic datasets vs real-world baselines.

User Requirements

High-fidelity metric extraction (mAP, FID), secure processing of sensor data using AES-256, and synchronized label alignment for seamless model training integration.

Technological Stack

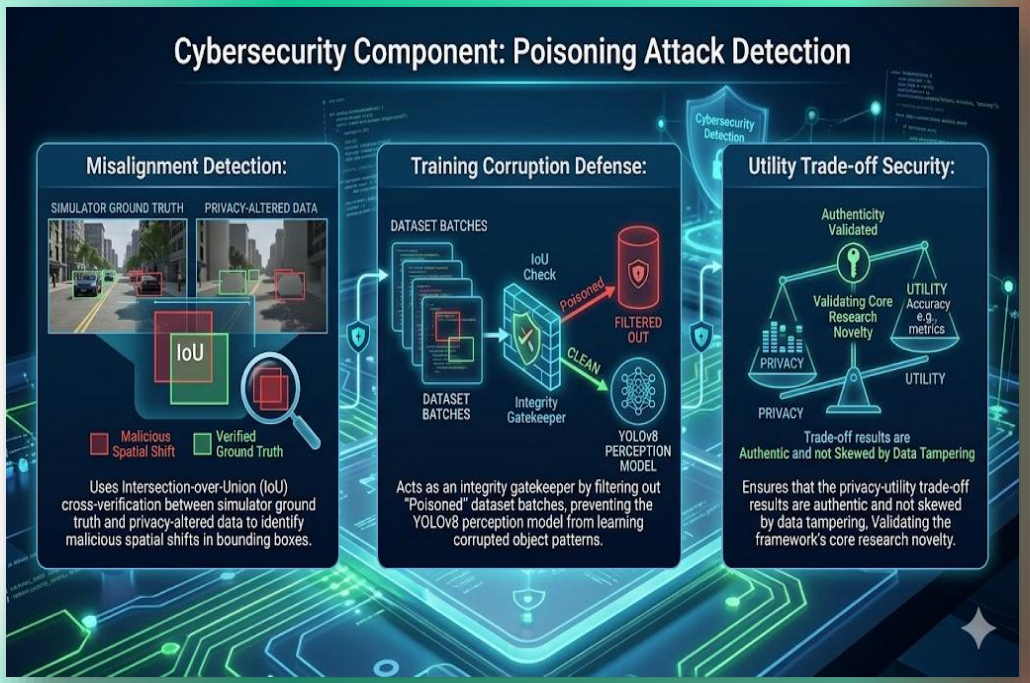
PyTorch, YOLOv8 for object detection, CARLA Simulator API, and PyCryptodome for high-grade cybersecurity decryption workflows.

CYBERSECURITY COMPONENT: POISONING ATTACK DETECTION

Misalignment Detection:
Uses Intersection-over-Union (IoU) cross-verification between simulator ground truth and privacy-altered data to identify malicious spatial shifts in bounding boxes.

Training Corruption Defense:
Acts as an integrity gatekeeper by filtering out "Poisoned" dataset batches, preventing the YOLOv8 perception model from learning corrupted object patterns.

Utility Trade-off Security:
Ensures that the privacy-utility trade-off results are authentic and not skewed by data tampering, Validating the framework's core research novelty.

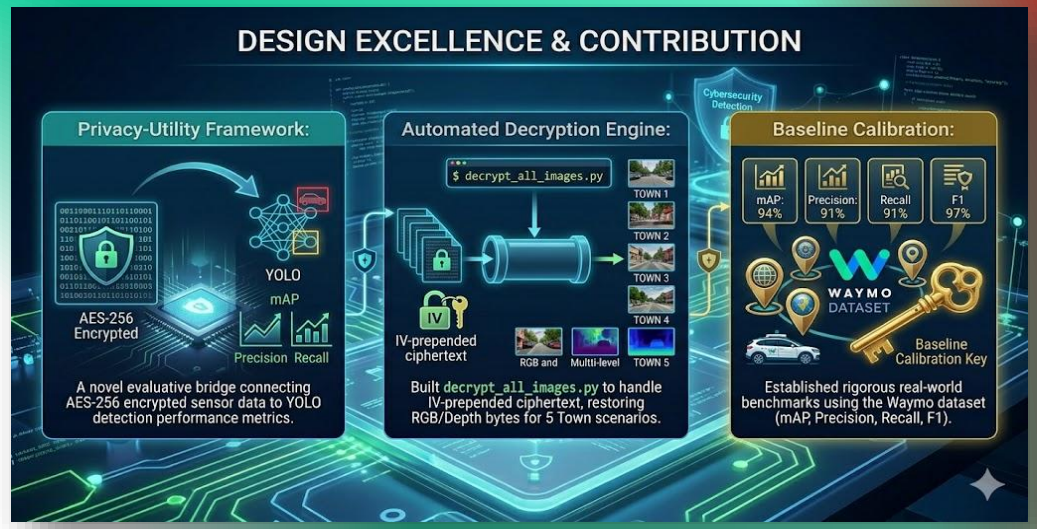


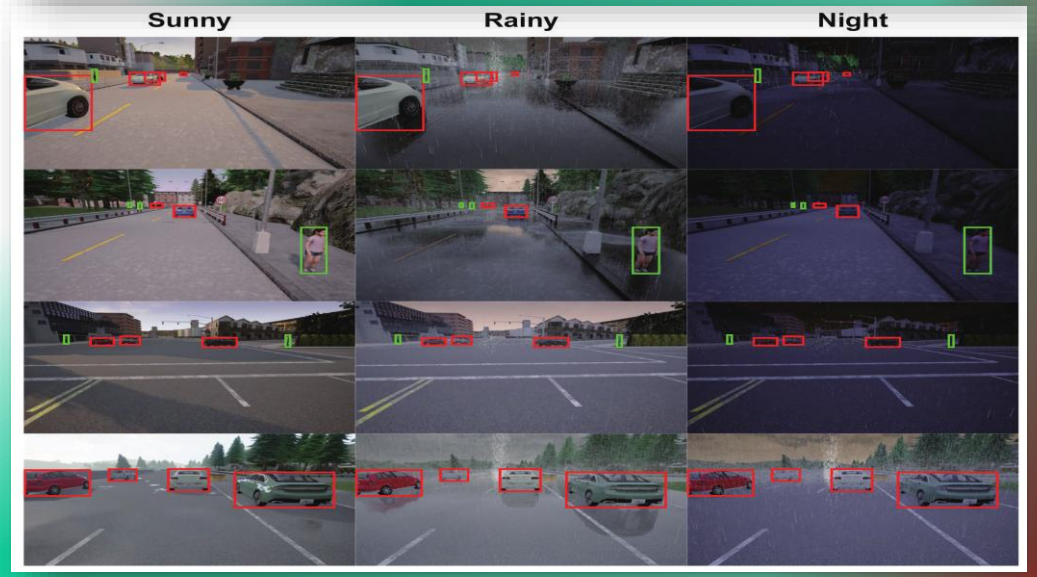
DESIGN EXCELLENCE & CONTRIBUTION

Privacy-Utility Framework:
A novel evaluative bridge connecting AES-256 encrypted sensor data to YOLO detection performance metrics.

Automated Decryption Engine:
Built decrypt_all_images.py to handle IV-prepended ciphertext, restoring RGB/Depth bytes for 5 Town scenarios.

Baseline Calibration:
Established rigorous real-world benchmarks using the Waymo dataset (mAP, Precision, Recall, F1).

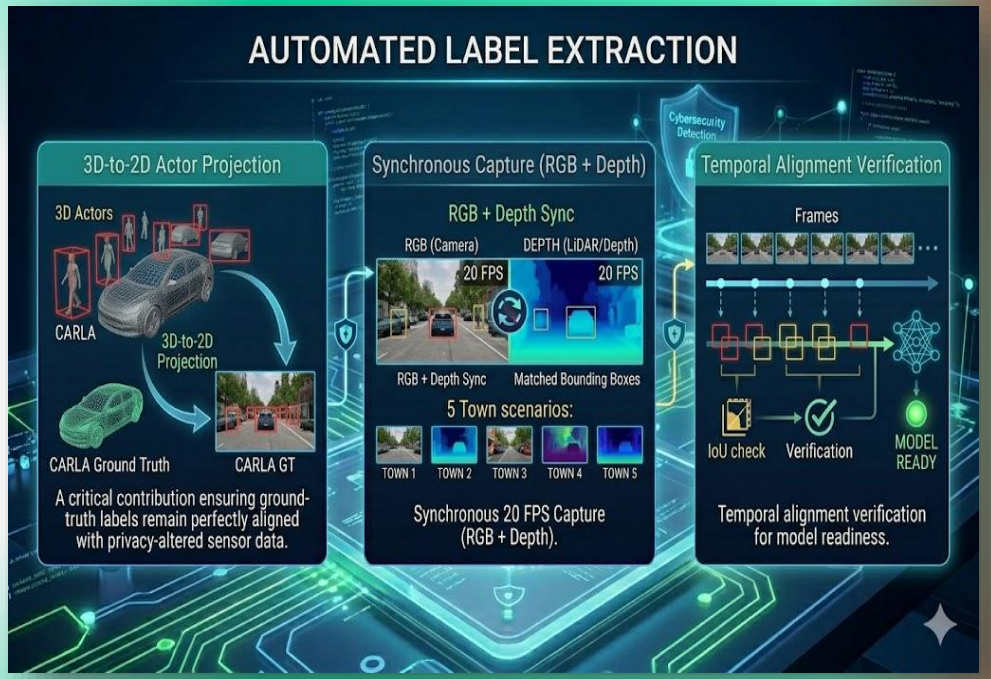




AUTOMATED LABEL EXTRACTION

A mission-critical contribution ensuring ground-truth labels remain perfectly aligned with privacy-altered sensor data.

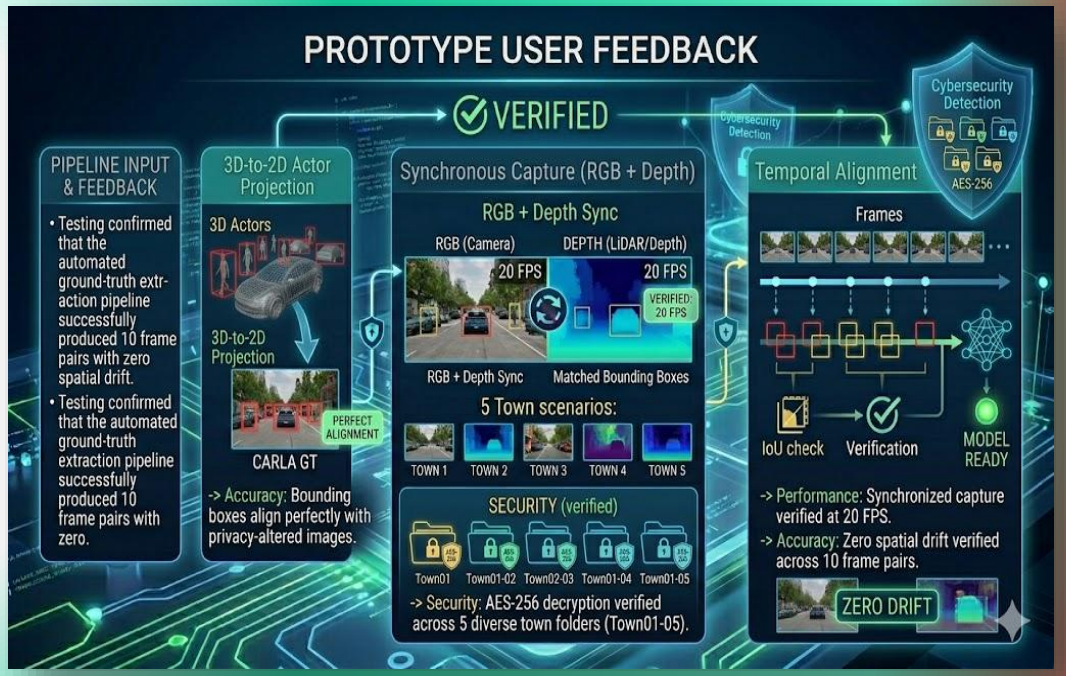
-> 3D-to-2D Actor Projection from CARLA Ground Truth.
-> Synchronous 20 FPS Capture (RGB + Depth).
-> Temporal alignment verification for model readiness.



PROTOTYPE USER FEEDBACK

Testing confirmed that the automated ground-truth extraction pipeline successfully produced **10 frame pairs** with zero spatial drift.

-> **Performance:** Synchronized capture verified at 20 FPS.
-> **Accuracy:** Bounding boxes align perfectly with privacy-altered images.
-> **Security:** AES-256 decryption verified across 5 diverse town folders (Town01-05).





IT22066916 | R.D.P.T. WIJESINGHE

Individual Component: **Risk and Compliance**

Novelty:

- Introduces an **attack-based privacy evaluation method** for synthetic AV data pipelines.
- Uses **Membership Inference Attacks (MIA)** to detect possible training data leakage.
- Provides **quantifiable privacy risk indicators**, enabling informed mitigation.

COMPLIANCE BASELINE

GDPR & NIST Privacy Framework mapped to R26-CS-010 (Phase 1)

GDPR	
Art. 5(1)(c)	Data Minimization
Art. 5(1)(f)	Integrity & Confidentiality
Art. 25	Privacy by Design
Art. 32	Security of Processing
Art. 35	Data Protection Impact Assessment
Art. 89	Research Safeguards

NIST Privacy Framework	
IDENTIFY	ID.RA - Risk Assessment ID.IM - Data Inventory & Mapping
GOVERN	GV.PO - Policies & Procedures GV.RM - Risk Management
CONTROL	CT.DM - Data Management CT.DP - Disassociation of Data
COMMUNICATE	CM.PO - Communication Policies
PROTECT	PR.DS - Data Security PR.PO - Protection Policies

THE PRIVACY PROBLEM I ADDRESS

Why privacy attack testing is critical to the whole project

The Challenge

- ! Even SYNTHETIC AV data can still leak real people's private information
- ! Faces, license plates and driving patterns can survive into synthetic outputs
- ! The team cannot CLAIM privacy without PROOF
- ! No standard way to test if a generative model memorized real training data
- ! GDPR Art. 25 requires privacy to be proven - not just assumed

My Solution

Attack 1 MIA	Test if the model memorized real training data using confidence score analysis
Attack 2 ReID	Test if synthetic data can be matched back to real individuals across 3 sensor types
Automated Runner	One command runs both attacks and generates a structured privacy report
GDPR Evidence	Reports map directly to GDPR articles - giving the team legal compliance evidence

Why It Matters

For the Project	Validates that the whole privacy framework actually works end to end
For IT22309556	My MIA results tell him if his differential privacy epsilon value is strong enough
For IT22541284	My ReID results confirm if her CARLA simulation is producing safe synthetic outputs
For IT22110220	My privacy reports give him the baseline to measure privacy vs utility trade-off
For GDPR	My test reports are the legal evidence that Art. 25 Privacy by Design is satisfied

My role = the **PROOF LAYER** of the project | If my attacks fail --> privacy is working | If they succeed --> team knows exactly where to fix

PRIVACY ATTACK TEST SCRIPTS

Developed test scripts for Membership Inference & Re-Identification attacks on synthetic AV data (Phase 2)

What I Built

mia_attack.py

Membership Inference Attack
Tests if model memorized real data
Using shadow model + AUC-ROC score

reid_attack.py

Re-Identification Attack
Tests if synthetic data matches real people
Tests Camera, LiDAR & Trajectory

run_all.py

Combined Test Runner
Runs both attacks automatically
Generates JSON privacy reports

How It Works

Real Calibration Data

↓

Trained AV Model

↓

MIA Attack
(Confidence Scores)

↓

ReID Attack
(Cosine Similarity)

↓

Privacy Report
(PASS / FAIL)

Team Integration

INPUT → IT22309556

Provides trained generator model
My MIA script tests whether
his model leaked real data

INPUT → IT22541284

Provides CARLA synthetic data
My ReID script tests whether
her data can re-identify people

OUTPUT → IT22110220

Uses my privacy reports
My results tell him where
privacy vs utility trade-off sits

Scripts are ready – waiting for real CARLA data and trained model to run final tests in Phase 3

Commercialization

Market Size

USD 556B

Global AV market projected by 2026 (Allied Market Research)

Primary: AV manufacturers and automotive OEMs under GDPR, CCPA, or PDPA needing compliant training data at scale.
Secondary: Academic research labs, regulatory bodies, and insurance telematics operators.

Customer Persona

ML Engineers and Data Infrastructure Leads
at AV development companies and research organisations.

Need: Train perception models at scale without triggering GDPR compliance reviews or expensive data licensing agreements.
Pain point: Real data carries legal risk. Existing synthetic tools carry no formal privacy proof.

What is Unique in Our Solution

No current platform demonstrates epsilon-DP compliance across both the calibration and training stages.
NVIDIA DRIVE Sim and Wayve produce high-fidelity outputs but carry no formal privacy certificate.

This system delivers:
Auditable epsilon-DP proof + GDPR-compliant dataset + MIA-validated privacy guarantee in one pipeline.

How Will the Cost Be Recovered

Total project cost: LKR 40,000 (AWS compute + storage). No software license fees

SaaS Model:
Clients submit calibration config and receive a certified synthetic dataset with epsilon-DP audit report. Per-run pricing and subscription tiers.

B2B Licensing:
Embed the full pipeline into CARLA or NVIDIA DRIVE Sim under an annual agreement. IP candidate for provisional patent.



Thank You!